

MPP-611AL

A highly engineered HDPE/aluminum oxide nanocomposite for unsurpassed scratch and scuff resistance

Features and Benefits

- Maximum scratch and scuff resistance with slip and lubricity
- HDPE composite reinforced with 300 nm aluminum oxide nanoparticles (Mohs Hardness 9)
- Easier-to-disperse nanotechnology in a safe, non-nano powder
- Composite particle density maximizes efficiency and minimizes settling
- Effective replacement for PTFE additives

Composition

HDPE/aluminum oxide

Recommended Addition Levels

0.5-1.5% (on total formula weight)

Systems and Applications

Water based, solvent based and energy curable coatings and inks. Industrial coatings (including plastic and metal); stains, sealers and varnishes; wood coatings; printing inks and OPV's (including flexo and gravure); powder coatings; coil coatings; rubber additives.

Typical Properties*

	<u>MPP-611AL</u>
Mean Particle Size (µm)	3.5 - 5.5
Maximum Particle Size (µm)	15.56
Melting Point °C	110 - 116
Density @ 25 °C (g/cc)	0.99
NPIRI Grind	1.0 - 2.0

This product is also available as a water based wax dispersion - Microspersion 611AL-50

Mar-26

MP-28AL Series

Value engineered synthetic wax/aluminum oxide nanocomposites for surface durability and lubricity

Features and Benefits

- Improves scratch, scuff and mar resistance
- Adds surface slip and lubricity
- Choose XF grade for maximum gloss retention
- Choose G grade for maximum gloss reduction and burnish resistance
- Biodegradable to OECD 302C test standard
- Not a microplastic per ECHA definitions

Composition

Synthetic wax/aluminum oxide

Recommended Addition Levels

1.0-3.0% (on total formula weight)

Systems and Applications

Water based, solvent based and energy curable coatings and inks, Industrial coatings (including plastic, metal and masonry); stains, sealers and varnishes; wood coatings; printing inks and OPV's (including flexo and gravure); powder coatings; coil coatings.

Typical Properties*

	<u>MP-28AL-XF</u>	<u>MP-28AL</u>	<u>MP-28AL-G</u>
Mean Particle Size (µm)	4.0 - 6.0	6.0 - 8.0	10.0 - 14.0
Maximum Particle Size (µm)	15.56	22.00	44.00
Melting Point °C	114 - 118	114 - 118	114 - 118
Density @ 25 °C (g/cc)	0.99	0.99	0.99
NPIRI Grind	1.0 - 2.0	1.5 - 3.0	N/A

Mar-26

Spherex™ Series

Hard, monosized spherical powders that improve surface durability with stain and moisture resistance

Features and Benefits

- Superior scratch and abrasion resistance with high coefficient of friction
- Improved stain and moisture resistance at dosages of 2.0 - 8.0%
- Spherex 200 lowers gloss and improves burnish resistance in architectural paints
- Spherex 200XF ideal for gloss reduction and surface protection in wood and leather coatings
- Improves edge coverage in powder coatings, especially on sharp corners and weld seams
- Outstanding UV stability

Composition

Acrylic

Recommended Addition Levels

0.5-8.0% (on total formula weight)

Systems and Applications

Water based, solvent based and energy curable coatings. Floor coatings. Powder coatings. Leather coatings. Architectural wall and trim paints, stains, sealers and varnishes. Interior and exterior can and container coatings. Coil coatings. Wood coatings. Printing inks and OPV's.

Typical Properties*

	<u>Spherex 200</u>	<u>Spherex 200XF</u>
Mean Particle Size (µm)	18.5 - 21.5	5.5 - 6.5
Maximum Particle Size (µm)	24.00	8.00
Degradation °C	> 260	> 260
Density @ 25 °C (g/cc)	1.20	1.20

Apr-26



PolyGlide 1226XF

A highly engineered HDPE/ceramic/nanoceramic composite for abrasion resistance and lubricity

Features and Benefits

- HDPE composite reinforced with hard, inert ceramic microspheres and nanoceramic platelets
- Improved Taber abrasion resistance when compared to PE/PTFE additives
- Provides slip and lubricity
- Ideal for can and container coatings; 21CFR 175.300 approved
- Effective replacement for PTFE additives
- Compare to (coarser) Polyfluor 900

Composition

HDPE/ceramic

Recommended Addition Levels

0.5-1.5% (on total formula weight)

Systems and Applications

Water based, solvent based and energy curable coatings and inks. Industrial coatings (including plastic and metal); stains, sealers and varnishes; wood coatings; printing inks and OPV's (including flexo and gravure); powder coatings; can, container, and coil coatings; rubber additives.

Typical Properties*

	<u>PolyGlide 1226XF</u>
Mean Particle Size (µm)	3.5 - 5.5
Maximum Particle Size (µm)	15.56
Melting Point °C	109 - 115
Density @ 25 °C (g/cc)	0.99
NPIRI Grind	1.0 - 2.0

This product is also available as a water based wax dispersion - Microspersion 1226XF-50

Mar-26

PolyTuf® 1229

A highly engineered LDPE/ceramic/nanoceramic composite for abrasion resistance and surface durability

Features and Benefits

- High performance additive for maximizing surface abrasion resistance (Taber) and film toughness
- Fortified with two types of hard, inert ceramic particles
- Low density polyethylene provides superior surface toughness, durability and mar resistance
- Ideal for walking surfaces
- PTFE-free alternative to Polyfluor 900
- Easy to disperse fine powder that can be incorporated with high speed mixing

Composition

Ceramic modified polyethylene

Recommended Addition Levels

0.5-2.0% (on total formula weight)

Systems and Applications

Water based, solvent based and energy curable coatings and inks. Industrial coatings (including plastic, metal and masonry); architectural wall and trim paints; stains, sealers and varnishes; floor coatings; wood coatings; printing inks and OPV's (including flexo and gravure); powder coatings; coil coatings; rubber additives.

Typical Properties*

	<u>PolyTuf 1229</u>
Mean Particle Size (µm)	9.0 - 12.0
Maximum Particle Size (µm)	31.11
Melting Point °C	110 - 113
Density @ 25 °C (g/cc)	0.97
NPIRI Grind	4.0 - 6.0

This product is also available as a water based wax dispersion - Microspersion 1229-40

Mar-26

MicroKlear 418AL

Aluminum oxide modified carnauba wax for maximum scratch resistance with surface slip, clarity and gloss retention. Also useful as an additive in seed coatings

Features and Benefits

- Carnauba wax composite reinforced with 300 nm aluminum oxide nanoparticles (Mohs Hardness 9)
- Provides maximum scratch resistance with surface lubricity
- Does not adversely affect clarity or gloss
- Easier-to-disperse nanotechnology in a safe, non-nano powder
- Composite wax/aluminum oxide particle is less abrasive on processing equipment compared to free aluminum oxide
- Effective replacement for PTFE additives

Composition

Carnauba wax/aluminum oxide

Renewable Carbon Index

>95%

Recommended Addition Levels

1.0-3.0% (on total formula weight)

Systems and Applications

Water based, solvent based and energy curable coatings and inks. Industrial coatings (including plastic, metal and leather); wood coatings; printing inks and OPV's (including flexo and gravure); interior and exterior can and container coatings; coil coatings; seed coatings.

Typical Properties*

	<u>MicroKlear 418AL</u>
Mean Particle Size (µm)	6.0 - 8.0
Maximum Particle Size (µm)	22.00
Melting Point °C	81 - 86
Density @ 25 °C (g/cc)	1.04
NPIRI Grind	2.0 - 3.5

This product is also available as a water based wax dispersion - Microspersion 418AL-40

Mar-26

MP-28AL Series

Value engineered synthetic wax/aluminum oxide nanocomposites for surface durability and lubricity

Features and Benefits

- Improves scratch, scuff and mar resistance
- Adds surface slip and lubricity
- Choose XF grade for maximum gloss retention
- Choose G grade for maximum gloss reduction and burnish resistance
- Biodegradable to OECD 302C test standard
- Not a microplastic per ECHA definitions

Composition

Synthetic wax/aluminum oxide

Recommended Addition Levels

1.0-3.0% (on total formula weight)

Systems and Applications

Water based, solvent based and energy curable coatings and inks, Industrial coatings (including plastic, metal and masonry); stains, sealers and varnishes; wood coatings; printing inks and OPV's (including flexo and gravure); powder coatings; coil coatings.

Typical Properties*

	<u>MP-28AL-XF</u>	<u>MP-28AL</u>	<u>MP-28AL-G</u>
Mean Particle Size (µm)	4.0 - 6.0	6.0 - 8.0	10.0 - 14.0
Maximum Particle Size (µm)	15.56	22.00	44.00
Melting Point °C	114 - 118	114 - 118	114 - 118
Density @ 25 °C (g/cc)	0.99	0.99	0.99
NPIRI Grind	1.0 - 2.0	1.5 - 3.0	N/A

Mar-26

MP-28AL Series

Value engineered synthetic wax/aluminum oxide nanocomposites for surface durability and lubricity

Features and Benefits

- Improves scratch, scuff and mar resistance
- Adds surface slip and lubricity
- Choose XF grade for maximum gloss retention
- Choose G grade for maximum gloss reduction and burnish resistance
- Biodegradable to OECD 302C test standard
- Not a microplastic per ECHA definitions

Composition

Synthetic wax/aluminum oxide

Recommended Addition Levels

1.0-3.0% (on total formula weight)

Systems and Applications

Water based, solvent based and energy curable coatings and inks, Industrial coatings (including plastic, metal and masonry); stains, sealers and varnishes; wood coatings; printing inks and OPV's (including flexo and gravure); powder coatings; coil coatings.

Typical Properties*

	<u>MP-28AL-XF</u>	<u>MP-28AL</u>	<u>MP-28AL-G</u>
Mean Particle Size (µm)	4.0 - 6.0	6.0 - 8.0	10.0 - 14.0
Maximum Particle Size (µm)	15.56	22.00	44.00
Melting Point °C	114 - 118	114 - 118	114 - 118
Density @ 25 °C (g/cc)	0.99	0.99	0.99
NPIRI Grind	1.0 - 2.0	1.5 - 3.0	N/A

Mar-26

SuperGlide 1231

A value engineered ceramic-modified amide wax for superior surface durability, lubricity, and antiblocking

Features and Benefits

- Amide wax composite reinforced with hard, inert ceramic microspheres
- Excellent scratch and abrasion resistance when compared to PTFE based additives
- Adds slip, release and lubricity with an excellent surface "feel"
- High melting point to improve heat and block resistance
- Lowers COF
- Not a microplastic per ECHA definitions

Composition

Ethylene bis(stearamide)/ceramic

Renewable Carbon Index

>80%

Recommended Addition Levels

0.5-3.0% (on total formula weight)

Systems and Applications

Water based, solvent based and energy curable coatings and inks. Wood coatings; printing inks and OPV's (including flexo and gravure); powder coatings; interior and exterior can and container coatings; powdered metals.

Typical Properties*

	<u>SuperGlide 1231</u>
Mean Particle Size (µm)	5.0 - 8.0
Maximum Particle Size (µm)	22.00
Melting Point °C	141 - 146
Density @ 25 °C (g/cc)	1.00
NPIRI Grind	2.0 - 3.5

Mar-26

Spherex™ Series

Hard, monosized spherical powders that improve surface durability with stain and moisture resistance

Features and Benefits

- Superior scratch and abrasion resistance with high coefficient of friction
- Improved stain and moisture resistance at dosages of 2.0 - 8.0%
- Spherex 200 lowers gloss and improves burnish resistance in architectural paints
- Spherex 200XF ideal for gloss reduction and surface protection in wood and leather coatings
- Improves edge coverage in powder coatings, especially on sharp corners and weld seams
- Outstanding UV stability

Composition

Acrylic

Recommended Addition Levels

0.5-8.0% (on total formula weight)

Systems and Applications

Water based, solvent based and energy curable coatings. Floor coatings. Powder coatings. Leather coatings. Architectural wall and trim paints, stains, sealers and varnishes. Interior and exterior can and container coatings. Coil coatings. Wood coatings. Printing inks and OPV's.

Typical Properties*

	<u>Spherex 200</u>	<u>Spherex 200XF</u>
Mean Particle Size (µm)	18.5 - 21.5	5.5 - 6.5
Maximum Particle Size (µm)	24.00	8.00
Degradation °C	> 260	> 260
Density @ 25 °C (g/cc)	1.20	1.20

Apr-26

MPP-620AL-XF

A highly engineered polyethylene/aluminum oxide nanocomposite that provides slip, abrasion/rub resistance, and antiblocking

Features and Benefits

- Maximum scratch and scuff resistance with slip and lubricity
- HDPE composite reinforced with 300 nm aluminum oxide nanoparticles (Mohs Hardness 9)
- Easier-to-disperse nanotechnology in a safe, non-nano powder
- Composite particle density maximizes efficiency and minimizes settling
- Effective replacement for PTFE additives

Composition

High density polyethylene/aluminum oxide

Recommended Addition Levels

0.5-1.5% (on total formula weight)

Systems and Applications

Water based, solvent based and energy curable coatings and inks. Industrial coatings (including plastic and metal); stains, sealers and varnishes; wood coatings; printing inks and OPV's (including flexo and gravure); powder coatings; coil coatings; rubber additives.

Typical Properties*

	<u>MPP-620AL-XF</u>
Mean Particle Size (µm)	3.5 - 5.5
Maximum Particle Size (µm)	15.56
Melting Point °C	120 - 124
Density @ 25 °C (g/cc)	1.00
NPIRI Grind	1.0 - 2.0

Mar-26

Superslip 6515AL

A highly engineered HDPE/EBS/aluminum oxide nanocomposite for maximum scratch, scuff and block resistance

Features and Benefits

- Maximum scratch resistance; equal to PE/PTFE additives
- HDPE/EBS composite reinforced with 300 nm aluminum oxide nanoparticles (Mohs Hardness 9)
- Good abrasion resistance with slip and lubricity
- Amide wax component provides excellent antiblocking and release properties
- Ideal for can and container coatings; 21CFR 175.300 approved

Composition

HDPE/EBS/aluminum oxide

Renewable Carbon Index

>90%

Recommended Addition Levels

0.5-1.5% (on total formula weight)

Systems and Applications

Water based, solvent based and energy curable coatings and inks. Industrial coatings (including plastic and metal); stains, sealers and varnishes; wood coatings; printing inks and OPV's (including flexo and gravure); powder coatings; can, container, and coil coatings; rubber additives.

Typical Properties*

	<u>Superslip 6515AL</u>
Mean Particle Size (µm)	3.5 - 5.5
Maximum Particle Size (µm)	15.56
Melting Point °C	139 - 145
Density @ 25 °C (g/cc)	0.99
NPIRI Grind	1.0 - 2.0

This product is also available as a water based wax dispersion - Microspersion 6515AL-40

Mar-26